

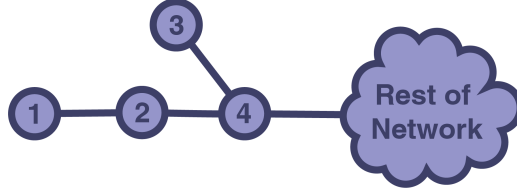
NETWORKS AND COMPLEXITY

Solution 20-4

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 20.4: From the tip of the tail [Lvl. 2]

Consider the following network:



- a) Suppose the adjacency matrix has an eigenvector $\mathbf{v}$ with eigenvalue λ . Normalize the first element of this eigenvector to $v_1 = 1$, then determine the eigenvector elements v_2 , v_3 and v_4 . For which values of λ is $v_3 > 0$?

Solution

We start from the eigenvector equation

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}. \quad (1)$$

Node 1 has only one neighbor, node 2, hence the first row of $\mathbf{A}$ only contains one non-zero element, $A_{1,2} = 1$. Hence, the first row of the eigenvector equation reads

$$A_{1,2}v_2 = \lambda v_1. \quad (2)$$

Using $A_{1,2} = 1$ we can write this as

$$v_2 = \lambda v_1 = \lambda. \quad (3)$$

The second row of the eigenvector equation has the form

$$A_{2,1}v_1 + A_{2,4}v_4 = \lambda v_2, \quad (4)$$

and hence

$$v_1 + v_4 = \lambda v_2 \quad (5)$$

Substituting $v_1 = 1$ and $v_2 = \lambda$ yields

$$1 + v_4 = \lambda^2 \quad (6)$$

and hence

$$v_4 = \lambda^2 - 1 \quad (7)$$

To find v_3 we consider the third row of the Eigenvector equation

$$A_{3,4}v_4 = \lambda v_3. \quad (8)$$

Solving for v_3 we find

$$v_3 = \frac{v_4}{\lambda} = \frac{\lambda^2 - 1}{\lambda} \quad (9)$$

The element v_3 is positive when

$$\frac{\lambda^2 - 1}{\lambda} > 0 \quad (10)$$

If λ is positive this yields the condition

$$\lambda > 1 \quad (11)$$

For the case $\lambda < 0$ we multiply both sides with λ ,

$$\lambda^2 - 1 < 0 \quad (12)$$

$$|\lambda| < 1 \quad (13)$$

Since we are looking for negative values of λ this yields the condition

$$-1 < \lambda < 0. \quad (14)$$

In summary, if $\lambda > 1$ then $v_3 > 0$ and in this case all of the elements of our four nodes are positive. The nodes in this sparsely connected branch find it difficult to sustain large eigenvalues, if other parts of the network want a large eigenvalue our four nodes have to pull all in the same direction to keep up.

If $0 < \lambda < 1$ then $v_3 < 0$. Checking the other nodes reveals that the sign change occurs between node 3 and node 4.

If $-1 < \lambda < 0$ then $v_3 > 0$, but $v_2 < 0$ and $v_4 < 0$.

If $\lambda < -1$ then $v_3 < 0$. Now a sign change occurs in every link of the branch. Since the branch is bipartite, alternating the signs in every step is a way to sustain relatively large negative eigenvalues.

- b) Consider also the case where instead of the branch a chain of 10 nodes is attached to the rest of the network. Compute the eigenvector elements for the nodes in the chain for the case $\lambda = 9$.

Solution

Let's number the nodes consecutively from the tip (node 1) to the end that is attached to the rest of the network (node 10). We start by choosing the normalization so that

$$v_1 = 1. \quad (15)$$

Node 1 only has one neighbor, node 2, and hence the first row of the eigenvector equation for this system reads

$$v_2 = \lambda v_1 = 9 \quad (16)$$

The eigenvector equation for the second row reads

$$v_1 + v_3 = \lambda v_2 \quad (17)$$

$$1 + v_3 = 81 \quad (18)$$

so

$$v_3 = 80 \quad (19)$$

From row three we get

$$v_2 + v_4 = \lambda v_3 \quad (20)$$

which we can write as

$$v_4 = 9v_3 - v_2 \quad (21)$$

and generally

$$v_{i+1} = 9v_i - v_{i-1} \quad (22)$$

so

$$v_4 = 9v_3 - v_2 = 720 - 9 = 711 \quad (23)$$

$$v_5 = 9v_4 - v_3 = 6399 - 80 = 6319 \quad (24)$$

$$v_6 = 9v_5 - v_4 = 56871 - 711 = 56160 \quad (25)$$

$$v_7 = 9v_6 - v_5 = 505440 - 6319 = 499121 \quad (26)$$

$$v_8 = 9v_7 - v_6 = 4492089 - 56160 = 4435929 \quad (27)$$

$$v_9 = 9v_8 - v_7 = 39923361 - 499121 = 39424240 \quad (28)$$

$$v_{10} = 9v_9 - v_8 = 354818160 - 4435929 = 350382231 \quad (29)$$

$$v_{11} = 9v_{10} - v_9 = 3153440079 - 39424240 = 3114015839 \quad (30)$$

I have gone to 11 (the first node in the rest of the network) because now we can normalize the eigenvector so that $v_{11} = 1$.

Using this normalization, we have

$$v_1 = 3.2112875e - 10 \quad (31)$$

$$v_2 = 2.89015871e - 9 \quad (32)$$

$$v_3 = 2.56902996e - 8 \quad (33)$$

$$v_4 = 2.28322538e - 7 \quad (34)$$

$$v_5 = 0.00000202921 \quad (35)$$

$$v_6 = 0.00001803459 \quad (36)$$

$$v_7 = 0.0001602821 \quad (37)$$

$$v_8 = 0.00142450431 \quad (38)$$

$$v_9 = 0.01266025673 \quad (39)$$

$$v_{10} = 0.1125178063 \quad (40)$$

$$v_{11} = 1.0 \quad (41)$$

So, the eigenvector stays very well confined to the rest of the network.